

Exercise 2

SQL AGGREGATE & OPERATIONS

① SELECT DISTINCT department
FROM student

Table: student

department

IT

HR

Finance

② SELECT department, AVG(age) AS Average-age
FROM students
GROUP BY departments

Table: student

department

average-age

IT

20,5

HR

22

Finance

23

③ SELECT *
FROM students
WHERE age BETWEEN 21 AND 23;

Student_id

Name

age

department

2

Bob

22

HR

3

Charlie

21

IT

4

Diana

23

Finance

5

Eve

22

HR

⑤ SELECT *

FROM Students

WHERE department IN ('IT', 'HR') AND age > 21;

Table: Student

Student-id	Name	age	department
2	Bob	22	HR
5	Eve	22	HR

⑥ SELECT department, sum(credits) AS total_credits

FROM courses

GROUP BY department

HAVING sum(credits) > 5

Table: Courses

department	total_credits
IT	11

⑦ SELECT *

FROM courses

WHERE credits < 4;

course-id	course-name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧ SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;

course_id	course_name	credits
102	Python	4
103	Data Science	4
104	SQL Basic	3

⑨ SELECT
MAX(grade) AS maximum_grade
MIN(grade) AS minimum_grade
AVG(grade) AS average_grade
FROM enrollments;

maximum_grade	minimum_grade	average_grade
90	78	84.6

⑩ SELECT course_id, COUNT(*) AS enrollment_count
FROM Enrollments
Group By course_id;

Table: Enrollments

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

① SELECT department

Sum(salary) AS total_salary

Sum(bonus) AS total_bonus

FROM salaries

GROUP BY department;

department	total_salary	total_bonus
IT	122 000	10 500
HR	109 000	7 500
Finance	70 000	6 000

② SELECT department, AVG(salary) AS average_salary

FROM employees

GROUP BY department

HAVING AVG(salary) > 55 000;

department	average_salary
IT	61 000
Finance	70 000

⑬ SELECT employee_id, name, salary, bonus,
 salary + bonus AS total_compensation
 FROM employees
 WHERE salary + bonus > 60 000

employee_id	name	salary	bonus	total_compensation
1	Tom	60 000	5 000	65 000
3	Spike	70 000	6 000	76 000
4	Tyke	67 500	5 500	73 000

⑭ SELECT department, SUM(budget) AS total_budget
 AVG(budget) AS avg_budget
 FROM Projects
 GROUP BY departments
 HAVING AVG(budget) > 70 000

department	total_budget	avg_budget
IT	170 000	135 000
Finance	80 000	80 000

⑮ SELECT project_id, project_name, department,
 budget
 FROM Projects
 WHERE budget BETWEEN 50 000 AND 120 000
 AND department <> 'Marketing';

Project_id	project_name	departments	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
3	Dashboard	IT	150 000
5	HR Portal	HR	50 000